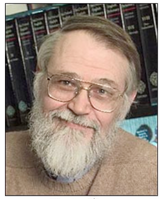
Advanced language features

- Macros and tricks for your instrument…

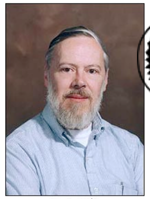
DECLARE / INITIALIZE / USERVARS %{%} sections

- Use the DECLARE section define user variables and functions.
  - **DECLARE** %{
  -    double some_global_var;
  - %}
- Use INITIALIZE for initialization of user variables and calculations.
  - **INITIALIZE** %{
  -    myvar = sqrt(PI*input_var)*rand01();
  - %}

K & R. / GNU



Brian Kernighan



Dennis Ritchie

- \- Both use normal c-syntax.

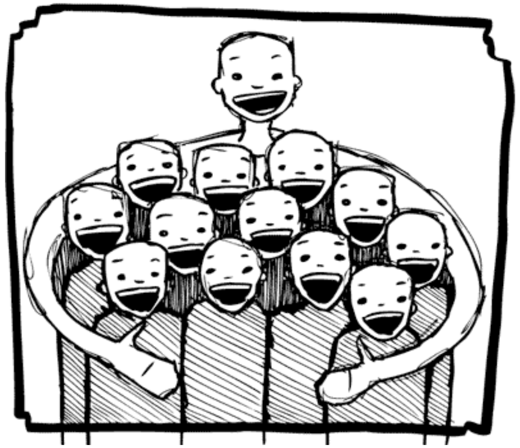
- Use USERVARS for particle-dependent flags **(McStas >=3 specific)**
  - **USERVARS** %{
  -    double myvar;
  - %}
- BEWARE: What you do in the c-style areas is c-standard, e.g. trigonometric functions from math.h use radians! - McStas ROTATED specifiers work in degrees, etc...

%include

- Instrumentfiles can include external c-code or other instrumentfiles... See the examples:

- ILL_H15_IN6.instr:%include "monitor_nd-lib"
- ILL_H16_IN5.instr:%include "ILL_H16.instr"
- ILL_H25_IN22.instr:%include "ILL_H25.instr"
- ILL_H25_IN22.instr:%include "templateTAS.instr"

- Used in the DECLARE / INITIALIZE / TRACE section



Instruments may depend on external commmands / linker instructions / searches etc.

- SHELL " some command run prior to code generation "
  - (No good example use at the moment)

- SEARCH "/add/a/component/search/directory/"
  - Used mainly in Greg Tucker's mcstas-antlr infrastructure for BIFROST that uses special repos

- SEARCH SHELL "wget https://some/component/file.comp"
  - Used mainly in Greg Tucker's mcstas-antlr infrastructure for BIFROST that uses special repos

- DEPENDENCY " -DSOME_DEFINE -llibrary -Iinclude "
  - Used e.g. for linking with MCPL and NCrystal libs, NeXus libs etc.
  - Raised both by comp and instrument level code

COMPONENT instance syntax in one, complex view…

{SPLIT} COMPONENT name = comp(parameters) {WHEN condition}
AT (...) [RELATIVE [reference|PREVIOUS] | ABSOLUTE]
{ROTATED {RELATIVE [reference|PREVIOUS] | ABSOLUTE} }
{GROUP group_name}
{EXTEND C_code}
{JUMP [reference|PREVIOUS|MYSELF|NEXT] [ITERATE number_of_times | WHEN condition] }

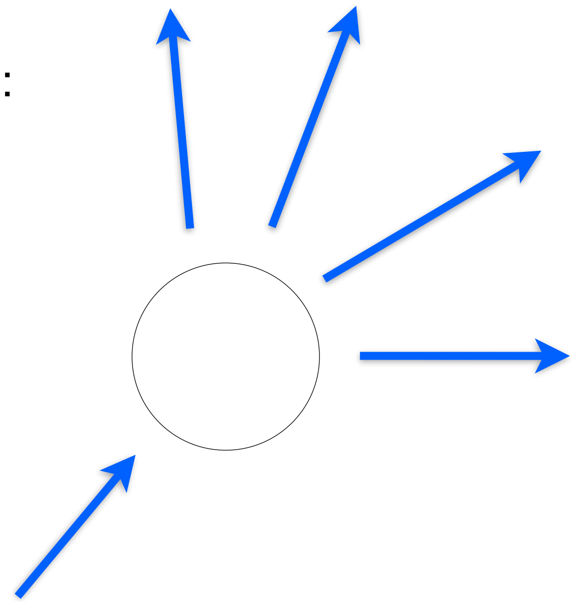
SPLIT

- Increase statistics beyond this point in the instrumentfile

- SPLIT n MyArm = Arm()
- AT somewhere

- will “formulate an if-statement”:

- for j=1:n
-   comp1
-   comp2
-   comp3
-   ...
- end (of instrument)

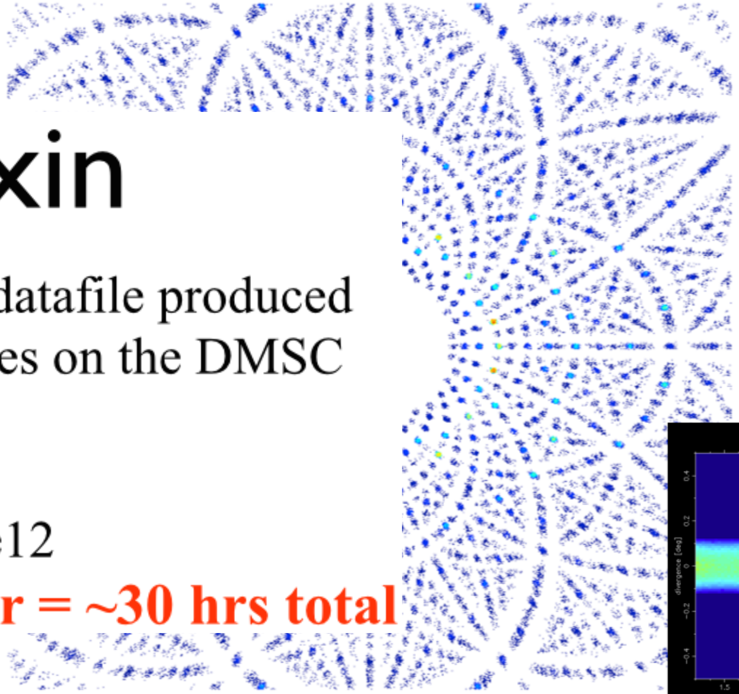


- ONLY meaningful in case of Monte Carlo choices after SPLIT point...

Problem: McStas Single_crystal.comp “slow” for large unit cell diffraction studies

- Example: Rubredoxin

1 timebin, 1000 x,y-bins



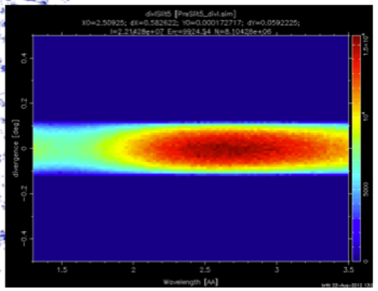
Rubredoxin

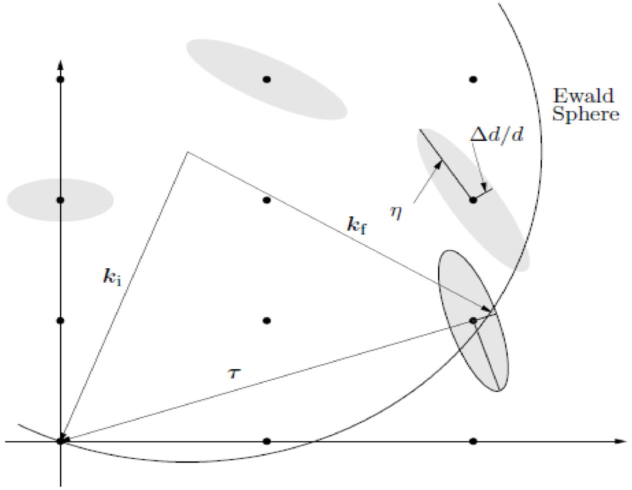
Images created from simulated datafile produced August 20th 2012 using 25 nodes on the DMSC cluster.

Neutron count: 1e12

Simulation time: ~10 + ~20 hr = ~30 hrs total

Neutroncount: 1e12
No gravitation
Xtal size: 0.5 mm
Xtal mosaicity: 12’
Detector: 50 x 50 cm flat
Detector-to-sample distance: 20 cm
Guide length: 131 m
Guide dimensions: 9.5 cm
λmin = 1.3 Å
λmax = 3.5 Å
Timespan: 51.39 to 143.4 ms
Divergence = 0.2 degs



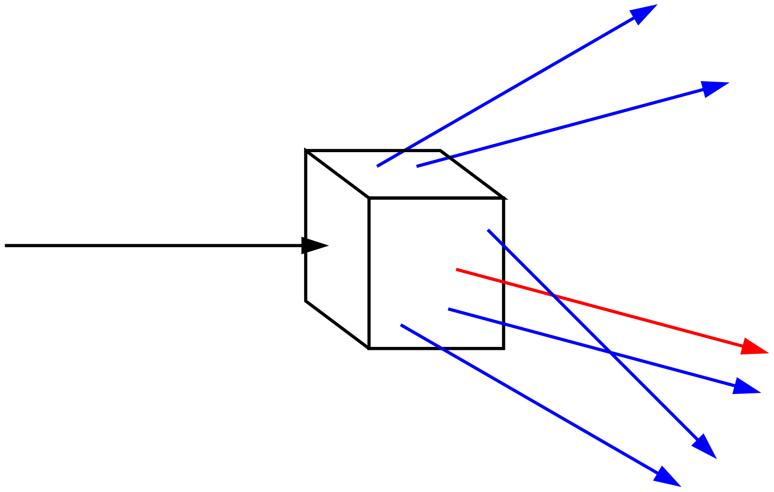


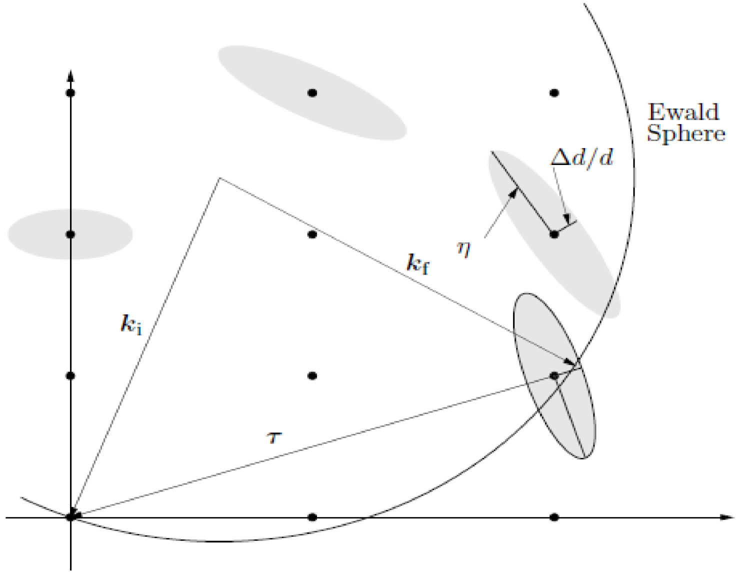
- Reflection list ~ 124 K reflections (still “small” in the PX world!!)

slight sidetrack….

Algorithm improvement: Use incoming neutrons more efficiently - scatter each one on all possible reflections

- **Red**: Original algorithm, one incoming neutron used only once
- **Blue**: Improved algorithm, each incoming neutron scattered (via SPLIT keyword) all possible times
- Component makes **estimate on average number of “active”** diffraction **spots** - in the case Rubredoxin this is around **50**!





!!! For now, does not work in our GPU implementation !!!

GROUP - components working in parallel

COMPONENT Mono1 = Monochromator_curved(...)

AT (0,0,-LMM) RELATIVE Cradle ROTATED (0,A1/2,0) RELATIVE Cradle

GROUP IN6Monoks

COMPONENT Mono2 = Monochromator_curved(...)

AT (0,0,0) RELATIVE Cradle ROTATED (0,A2/2,0) RELATIVE Cradle

GROUP IN6Monoks

- One comp after the other is "tried" in sequential order until the neutron was SCATTERED.

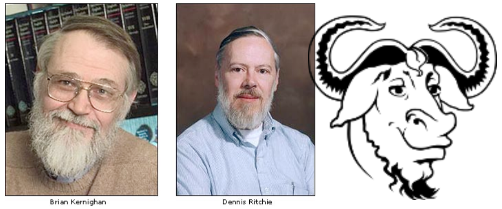
EXTEND

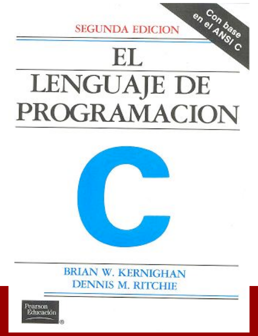
- Enrich component behaviour using EXTEND:

```
COMPONENT Mono1 = Monochromator_curved(...)
AT (0,0, -LMM) RELATIVE Cradle ROTATED (0,A1/2,0) RELATIVE Cradle
GROUP IN6Monoks
EXTEND
%{
  if (SCATTERED) { myvar = 1; }
%}

...
COMPONENT Mono2 = Monochromator_curved(...)
AT (0,0, 0) RELATIVE Cradle ROTATED (0,A2/2,0) RELATIVE Cradle
GROUP IN6Monoks
%{
  if (SCATTERED) { myvar = 2 ;}
%}
```

K & R. / GNU





WHEN

- Syntax:

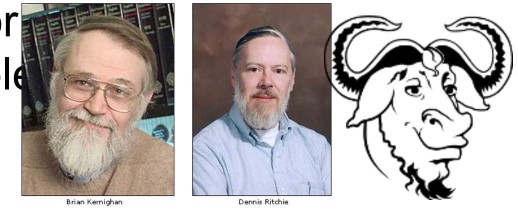
COMPONENT Mine = Yours(blah, blah)
WHEN (c-expression) AT (....)

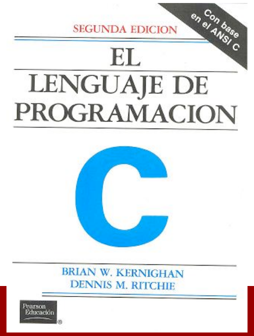
- Is very powerful when combined with EXTEND and user variables, or as a method to let input parameters select if certain components are active.

- Example: Use EXTEND to flag if neutron was scattered on one monochromator another. Then later use WHEN to only show contribution from blade N at sample

COMPONENT Mon = PSD_monitor(...)
WHEN (myvar==1) AT (0,0,0) RELATIVE Sample

K & R. / GNU





JUMP

- A goto. Be careful. Can be used in two situations:
- JUMP to myself
- JUMP to an Arm

- No coordinate transformations are applied... (Meaning that if the Arms you JUMP between do not coincide you will “move” / “reorient” the neutrons...)

- Syntaxes:
- COMPONENT a=b(...)
- WHEN (expr) AT (...) JUMP somewhere

- COMPONENT a=b(...)
- WHEN (expr) AT (...) JUMP myself

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Van Halen...

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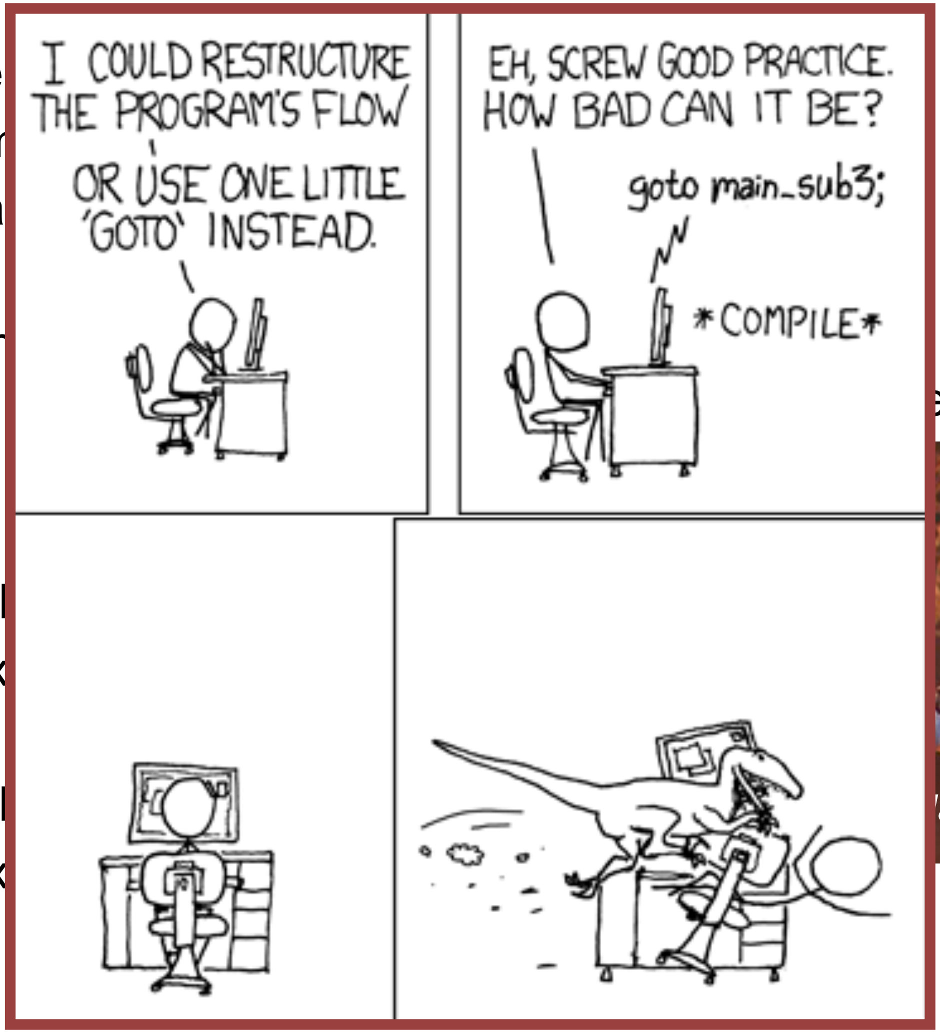
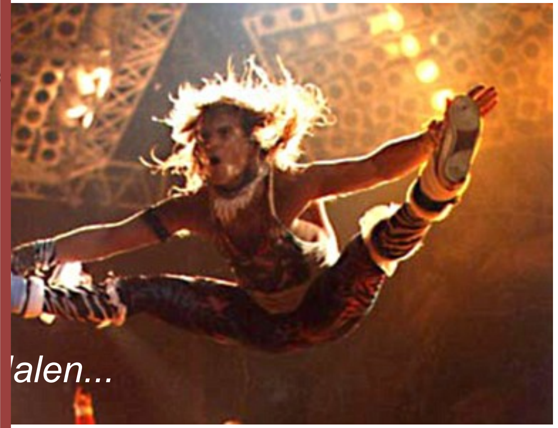


BEWARE - This IS a



/alen...

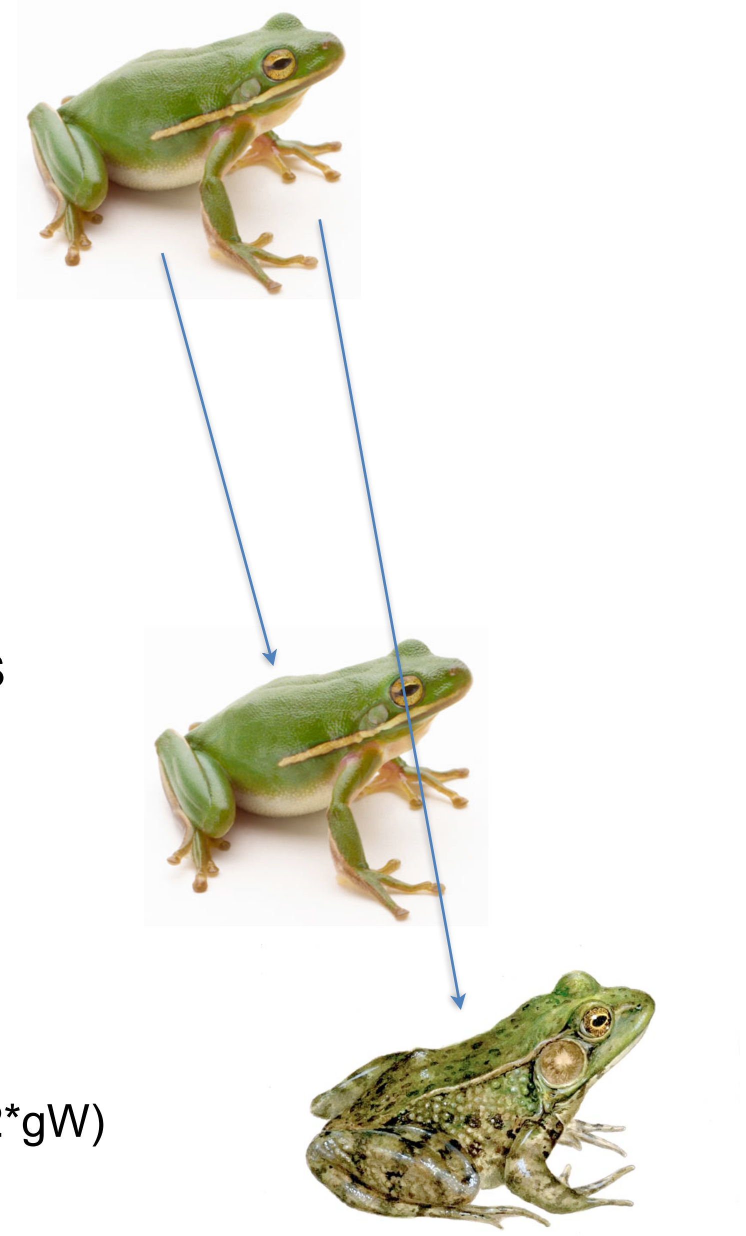
COPY- inside instruments

- In instruments: (see ILL_H25.instr)

- COMPONENT H25_1 = Guide_gravity(
- w1=0.03, h1=0.2, w2=0.03, h2=0.2, l=L_H25_1,
- R0=gR0, Qc=gQc, alpha=gAlpha, m=m, W=gW)
- AT (0,0,Al_Thickness+gGap) RELATIVE PREVIOUS
- ROTATED (0,Rh_H25_1,0) RELATIVE PREVIOUS

- COMPONENT COPY(H25_1) = COPY(H25_1)
- AT (0,0,L_H25_1+gGap) RELATIVE PREVIOUS
- ROTATED (0,Rh_H25_1,0) RELATIVE PREVIOUS

- COMPONENT COPY(H25_1) = COPY(H25_1)(W=2*gW)
- AT (0,0,L_H25_1+gGap) RELATIVE PREVIOUS
- ROTATED (0,Rh_H25_1,0) RELATIVE PREVIOUS



Example: Decompose multiple scattering from Single_crystal

```
USERVARS %{
   double multiple_scatt;
%}
…
COMPONENT Crystal = Single_crystal(… order=0 …)
AT (0,0,0) RELATIVE somewhere
EXTEND %{
   multiple_scatt=SCATTERED;
%}
…
COMPONENT PSD_single=PSD_monitor(…)
WHEN (multiple_scatt==1) AT (0,0,0) RELATIVE somewhere_else

COMPONENT PSD_multiple=PSD_monitor(…)
WHEN (multiple_scatt > 1) AT (0,0,0) RELATIVE somewhere_else
```

Or use with Monitor_nD user1="multiple_scatt"